

Mario Avolio

AI & CLOUD ENGINEER · GENERATIVE & AGENTIC AI

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Summary

AI & Cloud Engineer focused on production-grade generative AI: LLM applications, agentic workflows, and RAG pipelines, served as containerized microservices on the cloud. Research background in computer vision and model optimization, with a Best Paper Award at ICIAP 2025. I build reliable, evaluable systems that move from prototype to production without losing engineering rigor.

Experience

Machine Learning Reply

Milan, Italy

MACHINE LEARNING ENGINEER

Jan. 2026 - Present

- Build agentic workflows, RAG pipelines, and LLM-powered automation for enterprise use cases, from prototyping through deployment.
- Deploy and operate generative-AI solutions on cloud infrastructure with containerized services and CI/CD.
- Apply evaluation and observability practices to keep AI systems reliable once exposed to real users and data.

Imaging and Vision Laboratory (IVL), University of Milano-Bicocca

Milan, Italy

RESEARCH FELLOW

Jan. 2025 - Dec. 2025

- Built a large-scale synthetic Side-Scan Sonar dataset of 20,000+ labeled images for underwater mine detection, exceeding the scale of existing public datasets.
- Applied and optimized super-resolution models (SPAN, DRCT) to recover detection performance comparable to high-resolution data.
- Trained lightweight YOLOv11n/s detectors for robust detection in low-visibility conditions; work received a Best Paper Award at ICIAP 2025.

Skills

Generative AI & LLMs

LLM applications, RAG, agentic & multi-agent workflows, prompt & evaluation design, fine-tuning, vector databases, semantic search

Frameworks & Tooling

LangChain, PyTorch, Hugging Face, FastAPI, ChromaDB, OpenAI API, Pydantic

Cloud & MLOps

AWS, Docker, Docker Compose, CI/CD (GitHub Actions), microservices, model deployment & observability, Linux (Debian & RHEL)

Applied ML & CV

Image super-resolution, object detection & segmentation, model quantization, pruning, knowledge distillation

Programming

Python, SQL, Java, C++, JavaScript, R, Bash

Languages

Italian (native), English (fluent)

Education

University of Milano Bicocca

Milan (MI), Italy

M.S. IN COMPUTER SCIENCE (GRADE: 110/110 CUM LAUDE)

Sept. 2021 - March 2024

- **Thesis:** *Self-Supervised Learning and Model Adaptation for Facial Attribute Classification* (DINOv2, LoRA, PEViT).

University of Calabria

Arcavacata (CS), Italy

B.S. IN COMPUTER SCIENCE (GRADE: 108/110)

Sept. 2018 - Sept. 2021

Research Publications

Large-Scale Synthetic Side-Scan Sonar Dataset Generation and Super-Resolution Enhancement for Underwater Mine Detection [Best Paper Award]

ICIAP 2025, Rome, Italy

FIRST AUTHOR

Sept. 2025

From Vision to Execution: Enabling Knowledge Representation and Reasoning in Hybrid Intelligent Robots Playing Mobile Games

KR 2023, Rhodes, Greece

CO-AUTHOR

Sept. 2023

- Presented a hybrid AI framework integrating vision and symbolic reasoning for autonomous robotic gameplay.
- Demonstrated applications in match-3 and ball-sorting games using a delta robot with vision-based object recognition.